Mid Term Odd Semester Examination, Session 2021-22
M. Tech. Part I (Ist Semester) & Ph. D. Course Work
Mining Engineering
Subject: Geo-informatics, Subject Code: MN542

Time: 24 hours **Full Marks: 17**

Note: ***1. Answer all questions.***
2. Make all suitable assumptions wherever necessary.

1. Do as directed: **[0.5 X 4]**

(a) A digital image having a radiometric resolution of 6 bit, _____ is the maximum value and _____ is the minimum value of the DN could be represented by that image.

(b) State the name of any two sensors with its resolution.

(c) Differentiate between multispectral and hyperspectral sensors?

(d) If all visible wavelengths are reflected from an object, it will appear _____, while an object absorbing all visible wavelengths will appear _____.

2. **(a)** What type wavelength and sensor would be the best to monitor the general health of all vegetation cover over an area for several months, and why? **[1]**

(b) A company wants to start a surface mine in a mountainous region. Limiting geometric distortions as much as possible, would you choose a satellite-based or aircraft-based scanning system? Explain in terms of imaging geometry. **[2]**

3. **(a)** Why are most remote sensing systems avoided detecting and recording wavelengths in the ultraviolet and blue portions of the spectrum? **[1]**

(b) If the spectral range of the 400 channels of the CASI (Compact Airborne Spectrographic Imager) is exactly 0.40 μm to 0.90 μm and each band covers a wavelength of 1.25 nm will there be any overlap between the bands? **[2]**

4. **(a)** How does the EMR interact with; (i) The Earth (ii) The Atmosphere. **[1]**

(b) What do you understand by the term atmospheric window? Explain. **[2]**

5. **(a)** The infrared portion of the electromagnetic spectrum has two parts: the reflective and the emissive. Can you take photographs in these wavelength ranges? **[2]**

(b) Is there a passive equivalent to the radar sensor? If yes, then state its applications. **[1]**

6. **(a)** If you wanted to map the deciduous (e.g., maple, birch) and the coniferous (e.g., pine, fir, spruce) trees in a forest in summer using remote sensing data, what would be the best way to go about this and why? Use the reflectance curves illustrating the spectral response patterns of these two categories to help explain your answer. **[2]**

(b) What would be the advantage of displaying various wavelength ranges, or channels, in combination as colour images as opposed to examining each of the images individually? **[1]**

****